Assignment 4 Report :

What does Named Entity Recognition do?

NER is the process of automatically identifying and classifying key pieces of information (called "entities") in unstructured text into predefined categories.

Common entity categories include:

- **PERSON:** Names of people (e.g., *Alan Turing*, *Marie Curie*)
- **ORG (Organization):** Companies, agencies, institutions (e.g., *Google*, *UNESCO*)
- **GPE / LOC (Geopolitical Entity / Location):** Countries, cities, states, geographic features (e.g., *Milan*, *Mount Everest*)
- **DATE / TIME:** Specific days, years, or times (e.g., *June 11, 2026*)

Example

If you pass this sentence through an NER pipeline:

"Apple is looking at buying a startup in the United Kingdom for $1 billion."

An NER system like **spaCy** (which is mentioned in your task assignment) will detect:

- **Apple** $\rightarrow$ ORG
- **the United Kingdom** $\rightarrow$ GPE
- **$1 billion** $\rightarrow$ MONEY

What is GLiNER:

GLiNER stands for **Generalist and Lightweight Model for Named Entity Recognition**.

It is an open-source, highly efficient Natural Language Processing (NLP) framework designed to overcome the strict limitations of traditional NER tools (like standard spaCy models) and the high costs of Large Language Models (LLMs like ChatGPT).

Why is GLiNER a big deal?

To understand why GLiNER is unique, it helps to look at the two traditional ways people handled NER before it came along:

1. **Traditional Encoders (e.g., standard spaCy/BERT):** Fast and small, but locked into a **fixed set of labels** (like *PERSON, ORG, LOC*). If you suddenly want to

find *"Programming Language"* or *"Medical Symptom"*, you have to manually label thousands of sentences and completely retrain the model.
2. **Generative LLMs (e.g., ChatGPT/Gemini):** Excellent at finding any arbitrary entity type you ask for (**zero-shot capabilities**), but they are massive, slow, expensive to run via APIs, and require complex prompt-engineering.

GLiNER bridges this gap. It is a small, BERT-family model (often based on DeBERTa) that can extract **any** entity type you throw at it on the fly, without retraining, and it is lightweight enough to run quickly on standard CPUs.

How It Works: The "Matching" Architecture

Unlike generative LLMs that generate text word-by-word (which is slow), GLiNER treats NER as a **matching problem** in a shared space.

1. **The Input:** You give GLiNER the text *and* a list of custom labels you want to find.
    - *Example Text:* "We used Python to analyze data on the Apple MacBook."
    - *Example Labels:* ["Programming Language", "Hardware"]
2. **The Shared Latent Space:** GLiNER processes the text and your labels simultaneously. It projects the text chunks (spans) and the labels into a shared mathematical space.
3. **The Similarity Match:** It calculates a matching score between the vector of a word span (like *"Python"*) and the vector of a label (like *"Programming Language"*). If the score is high, it extracts it.

Key Advantages of GLiNER

- **Zero-Shot Flexibility:** You can define dynamic, highly specific categories ("Award", "Cryptocurrency", "Cuisine") on the fly, and it will extract them even if it has never seen that exact label during training.
- **Compact & Fast:** Models range from small (50M parameters) to large (300M parameters). They are 100x to 140x smaller than major LLMs, allowing them to run locally on consumer laptops or edge devices without needing a GPU.
- **Impressive Accuracy:** In benchmark tests, even the smaller versions of GLiNER regularly outperform ChatGPT and large instruction-tuned models at zero-shot entity extraction.
- **Supports Nested Entities:** Traditional NER struggles when one entity is inside another (e.g., in "University of Milan", "Milan" is a Location inside an Organization). GLiNER can handle these nested structures smoothly.

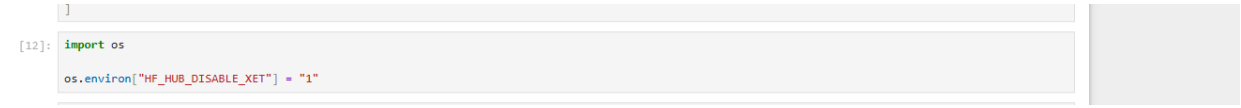
Expanding Ecosystem

Because of its popularity, the framework has expanded beyond its original scope:

- **spaCy Integration:** You can easily plug GLiNER directly into a spaCy pipeline using the gliner-spacy library.

- **GLiNER2 & Specialized Models:** The ecosystem includes specialized versions like **GLiNER-BioMed** (for complex medical terms) and **GLiNER-PII** (for redacting personal information locally). The updated **GLiNER2** framework unifies this with relation extraction (determining *how* two entities connect) and structured data extraction.

What does this code mean ?



```
]
[12]: import os

os.environ["HF_HUB_DISABLE_XET"] = "1"
```

This code is setting an **environment variable** in Python to change how Hugging Face (the platform where most modern NLP models like GLiNER, BERT, or Llama are stored) downloads files to your computer.

Report :

Assignment 4 – Named Entity Recognition and Linking with GLiNER and Wikipedia

Introduction

The objective of this assignment was to perform Named Entity Recognition (NER) and Named Entity Linking (NEL) on two domain-specific documents using the GLiNER model and Wikipedia. The first document focused on cybersecurity concepts, while the second document contained biomedical information. The assignment consisted of three exercises: entity extraction using GLiNER, entity linking using Wikipedia, and evaluation of the extracted entities using precision, recall, and F1-score.

Exercise 1 – Named Entity Recognition with GLiNER

In the first exercise, the GLiNER model was applied to two documents belonging to different domains: cybersecurity and biomedicine. Unlike traditional NER systems that rely on a fixed set of entity types, GLiNER allows users to define custom entity labels. This makes the model particularly useful for specialized domains.

For the cybersecurity document, the following labels were provided:

- vulnerability
- software
- protocol
- organization
- database
- programming language

The model analyzed the text and identified entity spans corresponding to these categories. Examples of extracted entities included:

- Heartbleed → vulnerability
- Rowhammer → vulnerability
- Spectre → vulnerability
- Log4Shell → vulnerability
- Apache Log4j library → software
- Java → programming language
- Google → organization
- National Vulnerability Database → database
- LDAP, RMI, and JNDI → protocol

For the biomedical document, the following labels were used:

- virus
- disease
- gene
- biological process
- technology

The model successfully recognized several biomedical entities, including:

- SARS-CoV-2 → virus
- COVID-19 → disease
- ACE2 → gene
- TMPRSS2 → gene
- infection susceptibility → biological process
- viral binding → biological process
- membrane fusion → biological process
- CRISPR-Cas9 → technology
- single-cell RNA sequencing → technology

After entity extraction, the identified entities were visualized directly within the original documents. This visualization allowed inspection of the detected entity spans and their assigned labels, confirming that GLiNER correctly identified most domain-specific entities.

The main purpose of this exercise was to understand how a modern NER system can identify relevant entities in specialized texts when provided with a user-defined set of labels.

Exercise 2 – Named Entity Linking with Wikipedia

The second exercise focused on Named Entity Linking (NEL). While NER identifies entity mentions and their types, NEL connects those mentions to specific real-world entities represented by knowledge bases such as Wikipedia.

The assignment provided a Wikipedia search function. For each entity detected by GLiNER, the system searched Wikipedia and returned the most relevant article title. The title was then converted into a Wikipedia URL.

For example:

- Heartbleed → https://en.wikipedia.org/wiki/Heartbleed
- Rowhammer → https://en.wikipedia.org/wiki/Row_hammer
- Spectre → https://en.wikipedia.org/wiki/Spectre_(security_vulnerability)
- Log4Shell → https://en.wikipedia.org/wiki/Log4Shell
- Google → https://en.wikipedia.org/wiki/Google

Similarly, biomedical entities were linked to Wikipedia pages:

- SARS-CoV-2 → Wikipedia article on SARS-CoV-2
- COVID-19 → Wikipedia article on COVID-19
- ACE2 → Angiotensin-converting enzyme 2
- TMPRSS2 → TMPRSS2
- CRISPR-Cas9 → CRISPR gene editing technology

Some entities did not receive valid links because Wikipedia search could not find a suitable page. Examples included abbreviations such as LDAP, JNDI, and RMI. In other cases, the returned result was not perfectly aligned with the intended meaning because the linking procedure relied solely on Wikipedia search without contextual disambiguation.

This exercise demonstrated the difference between entity recognition and entity linking. NER determines what the entity is and where it appears in the text, while NEL attempts to connect that entity to a specific knowledge-base entry.

Exercise 3 – Evaluation of Entity Extraction

The final exercise evaluated the quality of the entities extracted by GLiNER.

To perform the evaluation, manually curated gold-standard entity sets were created for both documents. These gold sets represented the entities considered correct and relevant according to human judgment.

For the cybersecurity document, the gold set contained entities such as:

- Heartbleed
- Rowhammer
- Spectre
- Log4Shell
- Apache Log4j
- Java
- LDAP
- RMI
- JNDI
- Google
- National Vulnerability Database

For the biomedical document, the gold set included:

- SARS-CoV-2
- COVID-19
- ACE2
- TMPRSS2
- infection susceptibility
- viral binding
- membrane fusion
- CRISPR-Cas9
- single-cell RNA sequencing
- disease progression

The entities predicted by GLiNER were extracted from the outputs generated in Exercise 1 and compared against these gold sets.

Three evaluation metrics were calculated:

Precision

Precision measures the proportion of predicted entities that are correct.

[
Precision = \frac{|Predicted \cap Gold|}{|Predicted|}
]

Recall

Recall measures the proportion of gold-standard entities that were successfully identified.

[
Recall = \frac{|Predicted \cap Gold|}{|Gold|}
]

F1-score

The F1-score combines precision and recall into a single metric.

[
F1 = \frac{2PR}{P+R}
]

The obtained results were:

Cybersecurity Document

- Precision = 0.692
- Recall = 0.818
- F1-score = 0.750

Biomedical Document

- Precision = 0.600
- Recall = 0.900
- F1-score = 0.720

The cybersecurity document achieved higher precision, indicating that most extracted entities were relevant. The biomedical document achieved higher recall, showing that GLiNER successfully identified most important biomedical entities, although several additional entities not included in the gold set reduced precision.

These results indicate that GLiNER performed well in both domains, particularly in terms of entity coverage, while occasionally producing extra entities that were not considered part of the gold standard.

Conclusion

This assignment demonstrated the complete workflow of information extraction using modern language models. First, GLiNER was used to perform named entity recognition on cybersecurity and biomedical texts. Second, the identified entities were linked to Wikipedia pages through a simple entity-linking pipeline. Finally, the extracted entities were evaluated using manually created gold-standard sets and standard evaluation metrics.

The results showed that GLiNER is capable of identifying domain-specific entities with good overall performance. The evaluation scores confirmed strong recall in both domains and reasonable precision, indicating that the model successfully captured most relevant entities while producing a limited number of false positives. Overall, the assignment provided practical experience with NER, NEL, visualization, and evaluation methodologies commonly used in information extraction systems.